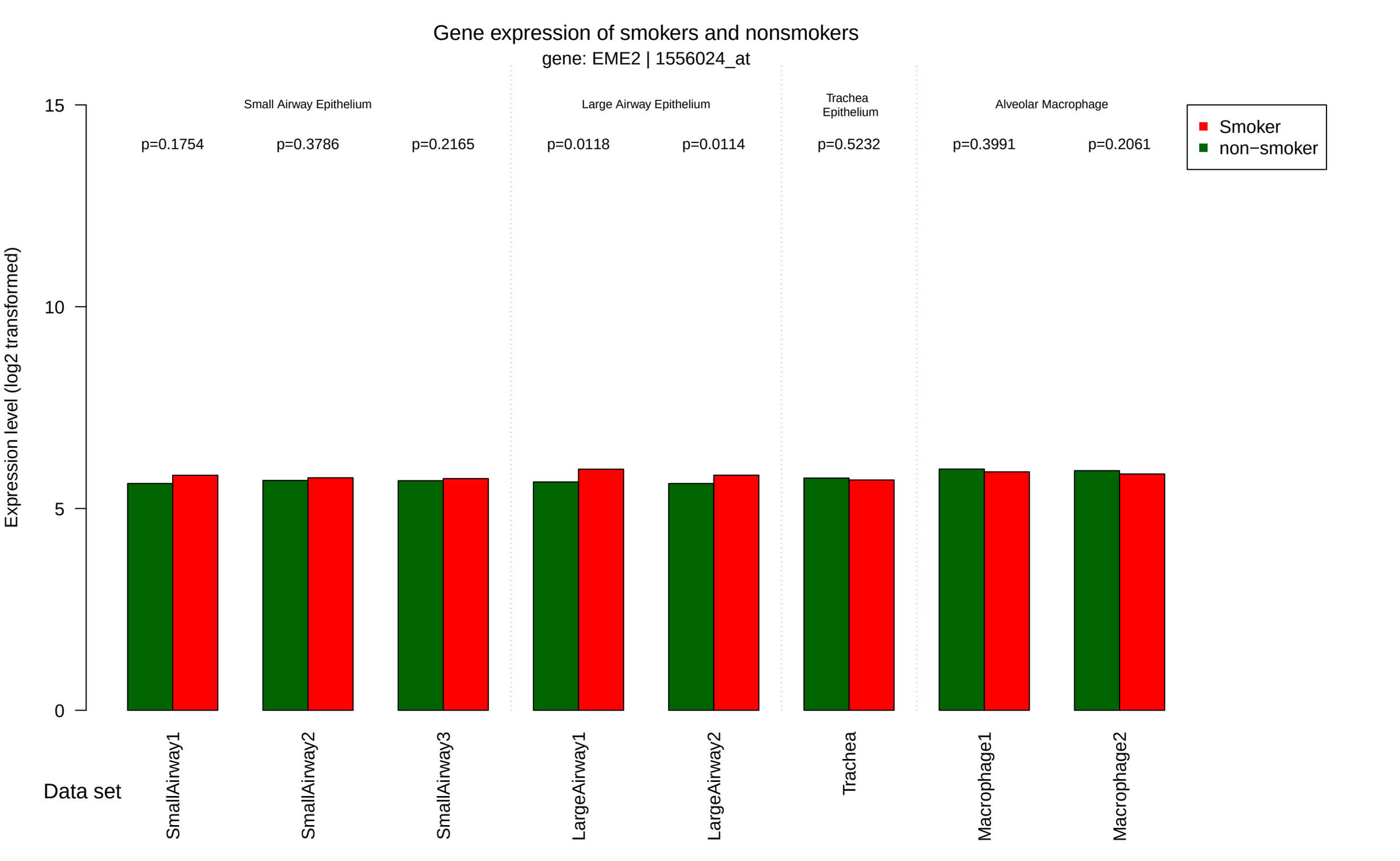
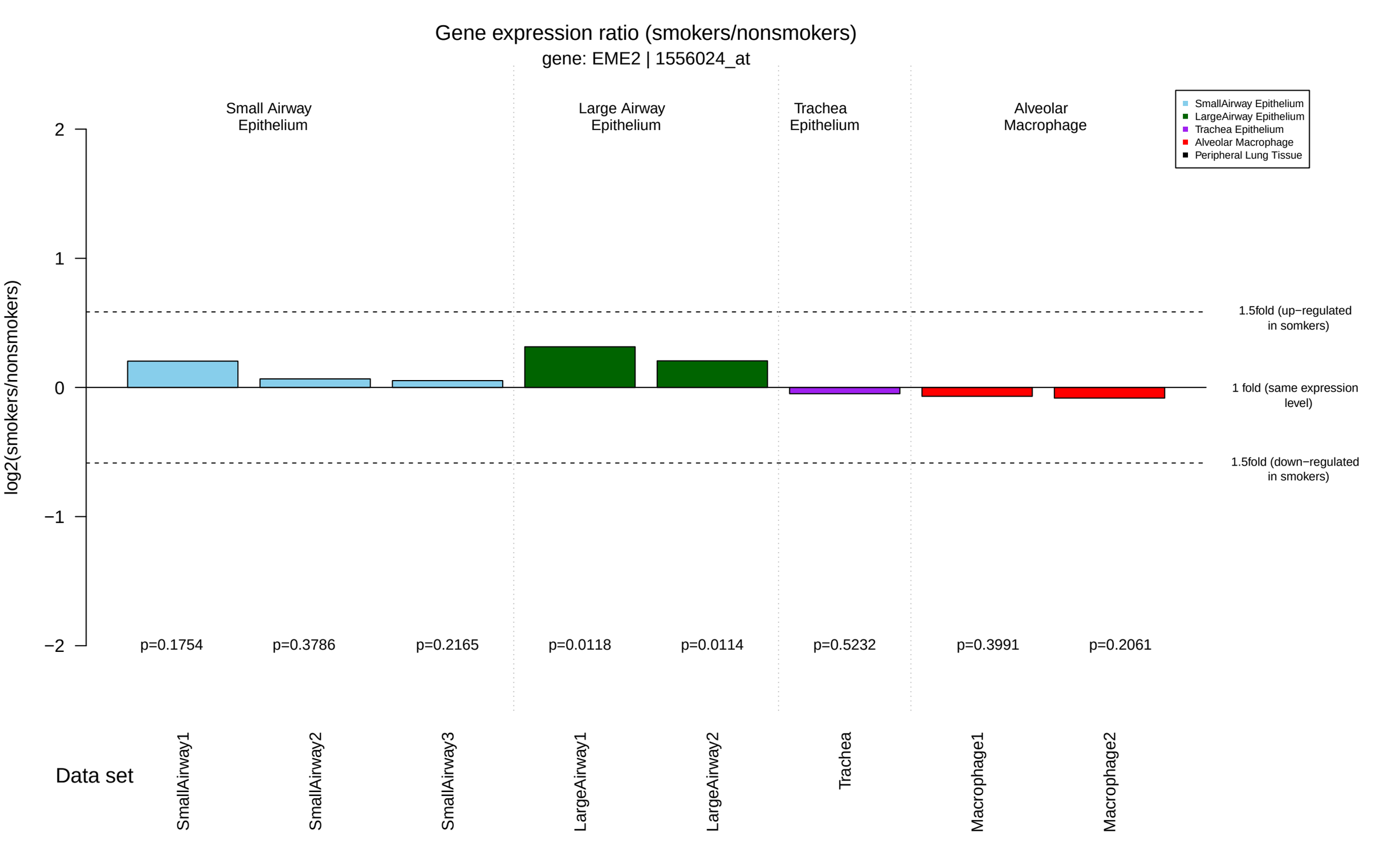
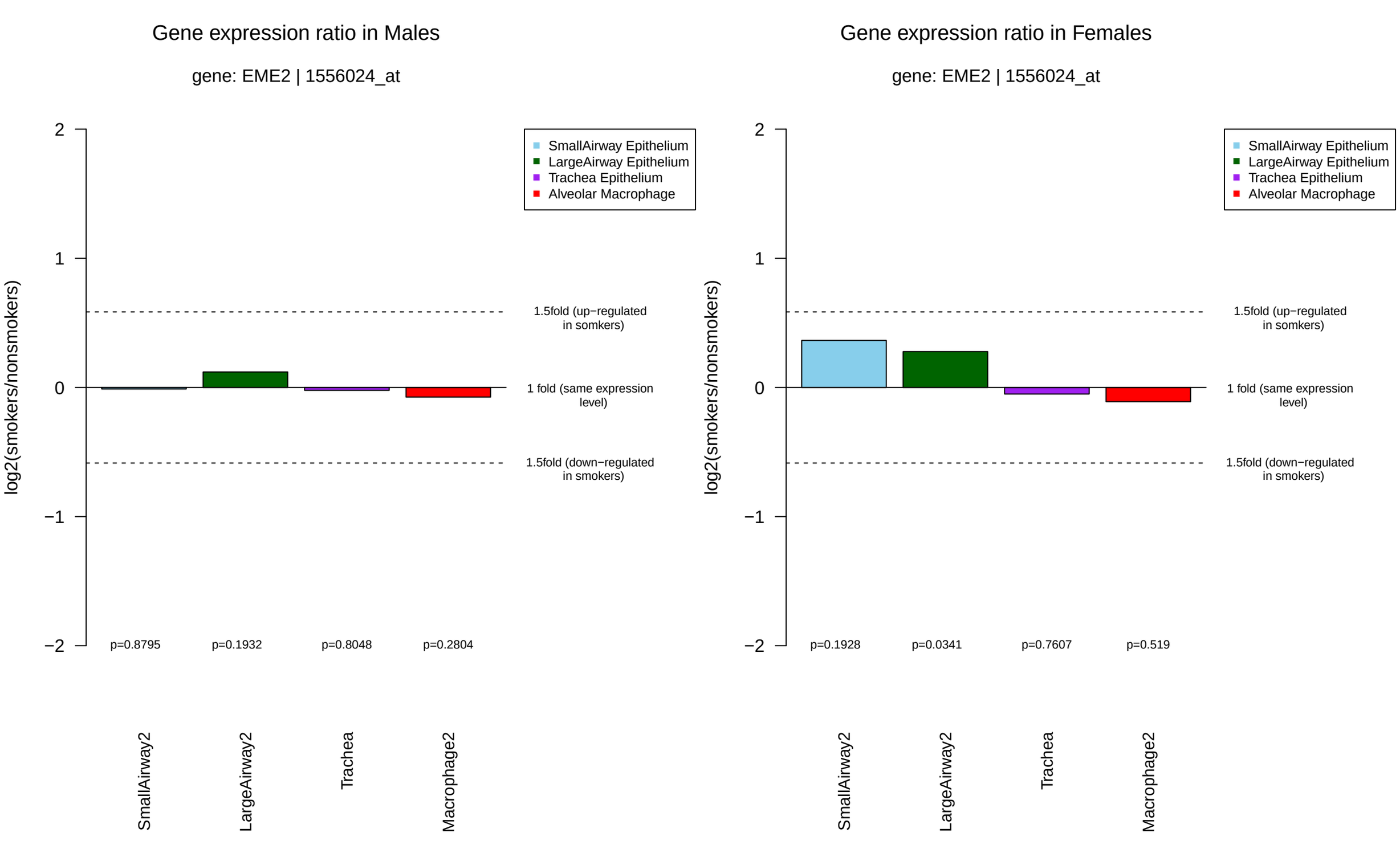
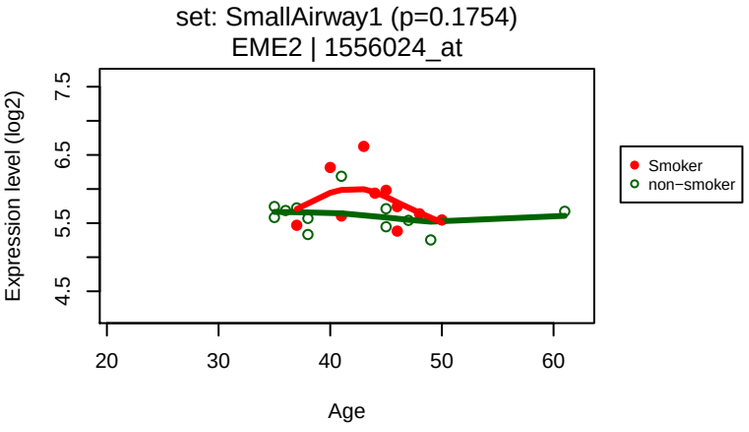
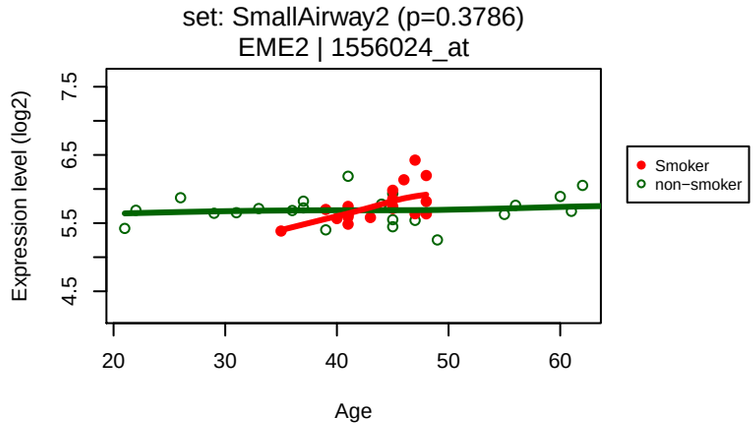
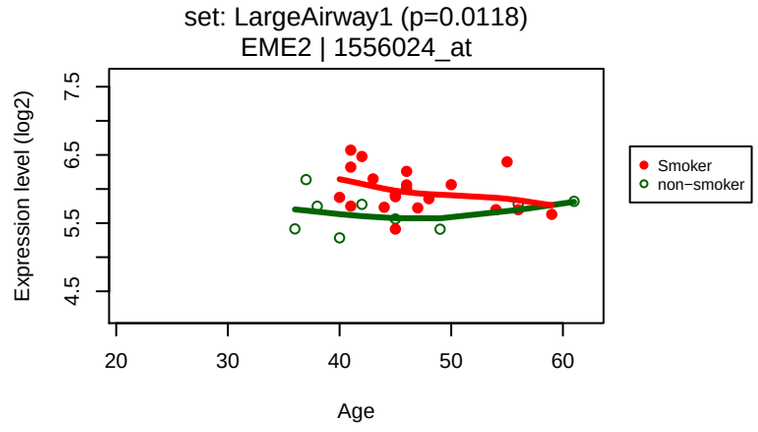

Gene expression of smokers and nonsmokers
gene: EME2 | 1556024_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1754
p=0.3786
p=0.2165
p=0.0118
p=0.0114
p=0.5232
p=0.3991
p=0.2061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

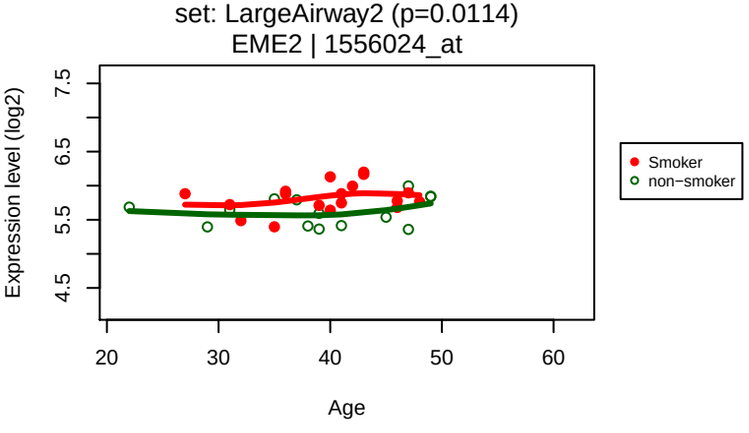

Gene expression ratio (smokers/nonsmokers)
gene: EME2 | 1556024_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1754
p=0.3786
p=0.2165
p=0.0118
p=0.0114
p=0.5232
p=0.3991
p=0.2061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

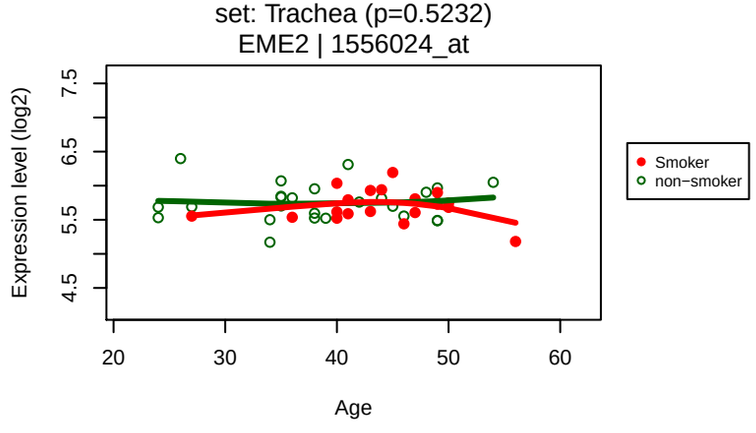

Gene expression ratio in Males
gene: EME2 | 1556024_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8795
p=0.1932
p=0.8048
p=0.2804
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EME2 | 1556024_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1928
p=0.0341
p=0.7607
p=0.519
SmallAirway2
LargeAirway2
Trachea
Macrophage2

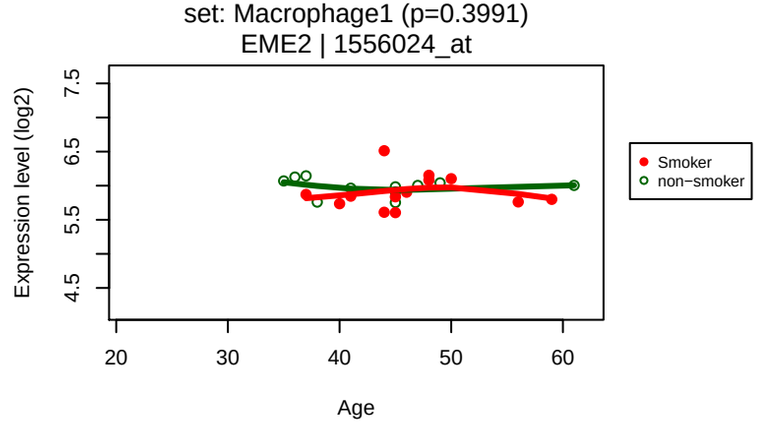

set: SmallAirway1 (p=0.1754)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

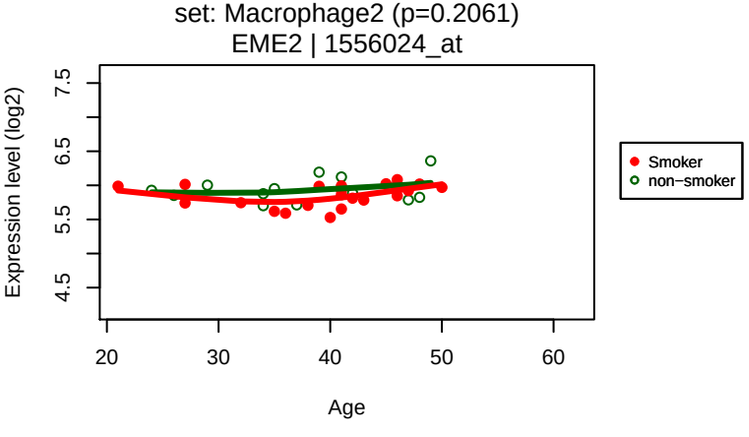

set: SmallAirway2 (p=0.3786)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

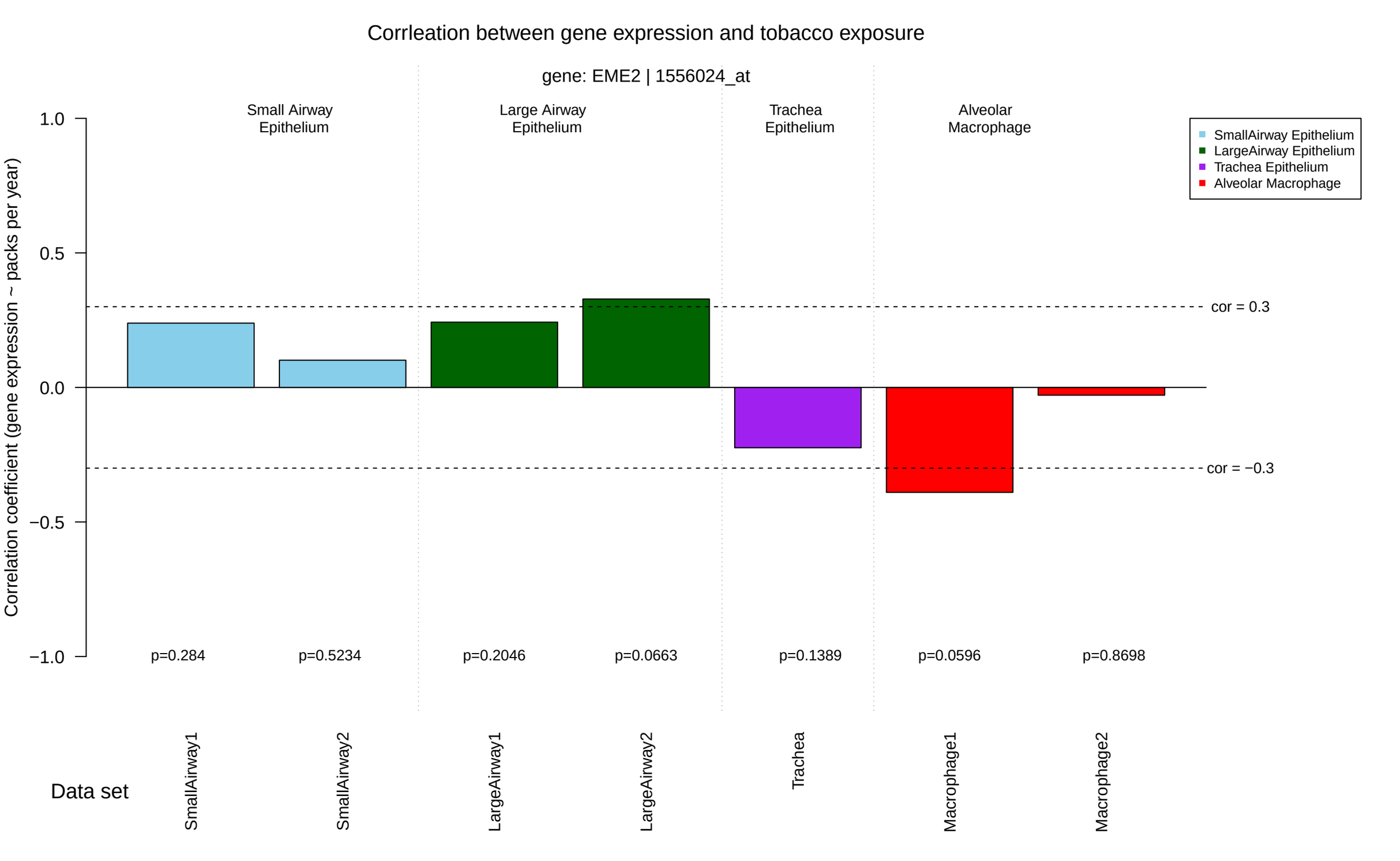

set: LargeAirway1 (p=0.0118)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0114)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5232)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3991)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2061)
EME2 | 1556024_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EME2 | 1556024_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.284
p=0.5234
p=0.2046
p=0.0663
p=0.1389
p=0.0596
p=0.8698
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2